

Flight-Control Forces, Moments, and Structural Load Paths

From Lilienthal's wing-tip devices to modern elevons

Name: _____ Team: _____

The question: When a control device changes an aerodynamic force, which aircraft moments change, and where does the structure receive the load?

Homework 1 integrated pressure around a two-dimensional airfoil. Here you integrate force along a wing, calculate moments about the center of gravity (CG), and examine how two elevons share pitch and roll commands. The same sequence applies throughout: **predict** → **sketch** → **compute** → **cross-check** → **explain**.

How to work. Use your existing three-person wing team to discuss figures and independently cross-check calculations. Rotate the roles of predictor, calculator, and checker. **Each student submits an individually authored report, MATLAB file, and Excel workbook.** Identify collaborators and explain any shared data-reading procedure. Include the individual reflection at the end of the handout. Due date and submission location are on Canvas.

Part	Task	Points
1	Read historical evidence; compare control mechanisms	15
2	Integrate spanwise force and moment increments in MATLAB and Excel	35
3	Calculate the moment of a side force at three locations	20
4	Mix elevon commands, impose limits, and calculate moments	20
5	Trace structural loads and calculate local string tension	10

Question points organize the technical tasks. Final evaluation follows the course homework rubric; the standard judgment record, one-slide brief, README and AI-use note are included in the submission instructions.

Download and open before starting

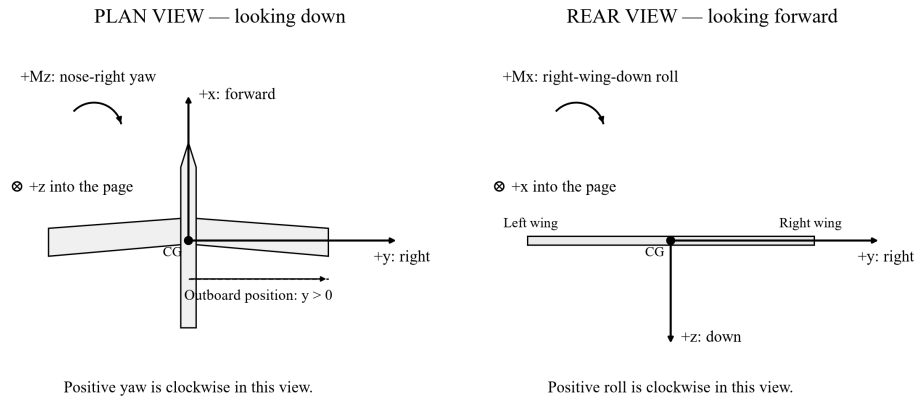
1. Open [Lecture 1 in Google Colab](#). Use **File** → **Save a copy in Drive** if you plan to change its controls.
2. Download [MIE446_HW2_Starter.m](#) and [MIE446_HW2_Excel_Starter.xlsx](#) into one HW2 folder. Keep original copies. Rename working copies with your surname.
3. Open the MATLAB file in the Editor. Set the Current Folder to your HW2 folder. Run the first setup section, then use **Run Section** for each numbered section. In MATLAB Online, upload the file first.
4. Open the Excel workbook and save your own copy. Cells designated for student work are intentionally blank. Enter **formulas**, not results copied from MATLAB.

The starter uses standard MATLAB functions and no add-on toolbox. NaN means “unfinished”; the original starter prints guidance instead of exporting an incomplete result. A file that runs with NaN entries is not a completed submission. The lecture notebook uses Python; your assessed computation here uses MATLAB and Excel.

Coordinate axes and moment reference points

Notebook route: find **2D. Aircraft rotations**, **2E. Center of gravity**, then **7A-3b. A vertical vane is not automatically a pure-yaw device**. Use Ctrl+F (Command+F on Mac) to find the section title. Read Figure 7A-3b-1 before the equations.

1. Declare the axes before drawing a moment



A circle with a cross means into the page. Right-handed body axes: x forward, y right, z down.

Lecture 1, Figure 7A-3b-1. Aircraft right means the pilot's right looking forward. An axis drawn into the page is not absent.

Use a right-handed body-axis system: **+x forward, +y right, +z down**. Positive M_x tends to lower the right wing; positive M_y tends to raise the nose; positive M_z tends to turn the nose right. A moment tendency does not by itself predict the complete flight trajectory.

$$\mathbf{M}_{CG} = \mathbf{r} \times \mathbf{F}, \quad \mathbf{r} = (x, y, z).$$

$$M_x = yF_z - zF_y, \quad M_y = zF_x - xF_z, \quad M_z = xF_y - yF_x.$$

The coordinates in $\mathbf{r}$ run **from the CG to the force application point**. Use metres and newtons, giving N·m. A positive coordinate locates a point; it is not automatically the perpendicular lever arm of every force component.

Do not carry coordinate signs between different problems without checking. HW1's airfoil coordinates had x from leading edge to trailing edge and y upward. Parts 2–4 here use body axes and moments about the CG. Part 5 instead supplies a local resisting torque about a pivot. The historical paper reports its own aerodynamic-axis coefficients; retain those labels and signs in Part 1. A lowercase roll coefficient c_l is not the lift coefficient C_L .

Practice example — different numbers from the assessed cases

A force $\mathbf{F} = (0, 0, 3)$ N acts at $\mathbf{r} = (0.20, 0, 0)$ m. Then $M_x = 0$, $M_y = -(0.20)(3) = -0.60$ N·m, and $M_z = 0$. Sketch why a downward force ahead of the CG produces a nose-down tendency.

```
r_demo = [0.20, 0, 0];    % row vector, m
F_demo = [0, 0, 3];       % row vector, N
M_demo = cross(r_demo, F_demo); % returns [Mx My Mz]
```

`cross(F_demo, r_demo)` gives the opposite sign. Use the physical definition $\mathbf{r} \times \mathbf{F}$, not whichever order produces the expected answer.

Read the device, then read the measurement

Notebook route: 7A-3 (device names), 7A-3a (Lilienthal's tail), 7A-3b (moments), and 7A-7–7A-13 (modern UAV examples). Open Raffel et al., “Flight Controls of Otto Lilienthal's Experimental Monoplane from 1895”. Read the device description and **Section II.C, Controllability**; inspect **Figures 11 and 12**. You need not summarize the whole paper.

1A. Identify the hardware [4 points]

1. Choose one historical photograph or the lecture's labeled conceptual plan view. Record its source. Label the **aft horizontal tail**, **aft vertical tail**, **wing-tip vane**, and one **control-linkage route**. If a component is hidden in your photograph, say “not visible in this view” and add a separate labeled sketch.
2. Write two or three sentences distinguishing **tail presence**, **surface motion**, and **control purpose**. Explain why the existence of a wing-tip rudder does not imply absence of an aft tail.

1B. Extract and interpret six pairs of coefficients [7 points]

1. In Figure 11, locate the roll-coefficient axis and the two marker series: **circles: spoileron, 90°**; **crosses: wing warping**. In Figure 12, find the corresponding yaw-coefficient series. **Do not use the square rudder markers** in this comparison.
2. At $\alpha = 6^\circ, 10^\circ, 14^\circ$, read the marker center for each mechanism. Use the nearest grid ticks to estimate the value. Read the **measured markers**, not the dashed fitted/trend line. Keep the plotted sign. Report sensible precision: graph reading does not justify many decimals.
3. Enter the six rows in the Excel **Paper** sheet, rows **7:12**. Columns A/B identify angle and mechanism; enter roll coefficient in C, yaw coefficient in D, reading tolerance in E, and figure/marker notes in F. In E, write separate roll and yaw tolerances if their plot scales differ, for example “roll $\pm \dots$; yaw $\pm \dots$ ”.
4. Use grid spacing and marker width to justify your tolerance. Compare one row with a teammate's independent reading. If values differ by more than the declared reading tolerance, revisit the figure together and record the resolution.
5. Explain the observed difference between the two mechanisms' roll–yaw coupling over these selected points. Cite the figure and test configuration. Do not claim a universal response at every angle of attack or every spoileron deflection.

Keep these coefficients nondimensional. Record the paper's coefficient labels and axis convention. No conversion to N·m is required; such a conversion would require its reference dimensions and coefficient definitions.

1C. Compare two modern examples [4 points]

Choose **one of MQ-9 or X-48B** and **one of X-36, X-47B, or MAGMA** from the notebook. Make a two-row table with: aircraft; visible tail/canard/winglet arrangement; one **documented** control mechanism; physical quantity it changes; and structure receiving the reaction. Cite the relevant notebook section and its linked source. If a photograph alone cannot establish a mechanism, state that limitation instead of guessing.

Submit for Part 1: one annotated figure; the six-row coefficient table with tolerances; a short coupling explanation; and the two-aircraft comparison. Aim for about one to two report pages.

Define the model before entering equations

Notebook route: 5. From nondimensional coefficients to forces and moment for dynamic pressure; **7A-3b** and its Figure 7A-3b-2 for signs; **7A-5. Predict the forces; audit the load path** for the physical mechanism.

Input	Value	Input	Value
Density ρ	1.225 kg/m ³	Baseline speed V	15 m/s
Half-span $s_{\max}$	0.450 m	Activation station a	0.300 m
Root chord c_r	0.160 m	Tip chord c_t	0.100 m
Lift-loss factor A_z	0.30	Drag-increase factor A_x	0.10

Let s measure distance along the **right** half-wing. Use a linearly tapered chord and an activation function that grows from zero at a to one at the tip:

$$c(s) = c_r + (c_t - c_r) \frac{s}{s_{\max}}, \quad q = \frac{1}{2} \rho V^2,$$

$$g(s) = \begin{cases} 0, & 0 \leq s < a, \\ \frac{s - a}{s_{\max} - a}, & a \leq s \leq s_{\max}. \end{cases}$$

The supplied **force increments per unit span** are

$$\Delta f_z(s) = q c(s) A_z g(s), \quad \Delta f_x(s) = -q c(s) A_x g(s).$$

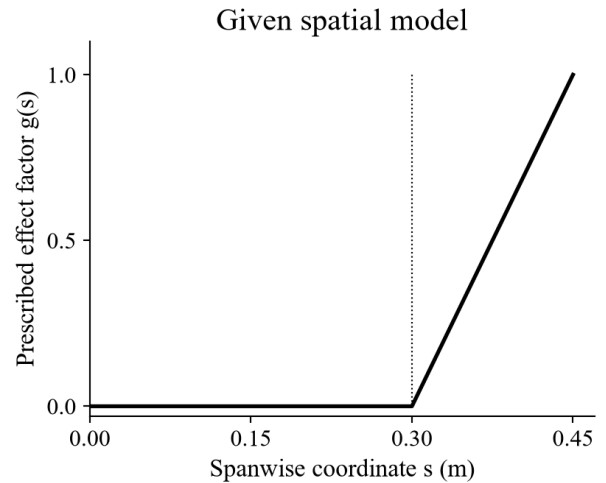
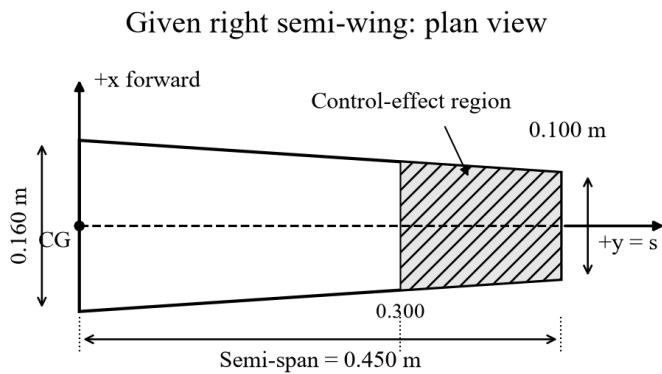
They have units N/m, because Pa $\times$ m = N/m. Here $\Delta f_z > 0$ means **loss of upward lift relative to the baseline**; it does not mean the complete wing lift acts downward. The added drag points aft, so $\Delta f_x < 0$.

Model boundary. These prescribed laws are an educational model, not measured Lilienthal data. For the moment calculation, idealize every force application point as $\mathbf{r}(s) = (0, s, 0)$ relative to the CG. The model dimensions resemble a small printed wing, but this is not a strength validation, a wing-root reaction calculation, or a prediction of its flight performance.

Excel input-cell map — WingLoads sheet

Input cell	Meaning	Input cell	Meaning
B5	ρ	B6	V
B7	$s_{\max}$	B8 / B9	Root / tip chord
B10	Activation a	B11 / B12	A_z / A_x
B13	Intervals, 90	B14	Dynamic pressure q

Locate the active region and the force line



Dashed force-reference line: $\mathbf{r} = (0, s, 0)$. This is an instructional geometry, not a flight-qualified design.

Instructional geometry. The application line $\mathbf{r} = (0, s, 0)$ is a declared force-reference idealization; it does not specify the actual aerodynamic center of a real wing.

Predict, then write the four integrals

Sketch the right half-wing and CG in plan and rear views. Mark the active region and both force increments. Predict the signs of all four results before running code:

$$\begin{aligned}\Delta F_z &= \int_0^{s_{\max}} \Delta f_z ds, & \Delta F_x &= \int_0^{s_{\max}} \Delta f_x ds, \\ \Delta M_x &= \int_0^{s_{\max}} s \Delta f_z ds, & \Delta M_z &= - \int_0^{s_{\max}} s \Delta f_x ds.\end{aligned}$$

Explain why the force integrals have units N and the moment integrals N·m. No derivation of the cross-product formula or exact antiderivative is required.

Write the prediction before calculating: which wing tends to move down, and which way does the nose tend to turn? Explain which force component creates each tendency. Then use the formulas to test your reasoning.

Optional visual check: point to the first station at which g is nonzero and compare the shapes you expect for chord and load. The activation function grows outward while the chord decreases; the load is their product, not either curve alone.

Build arrays, then integrate them

Open **Sections 2A–2C** of the starter. Replace each complete right-hand side marked **TODO**; leave the supplied variable name in place. Run the setup and Section 2A before Section 2B.

1. **Create the station grid.** The supplied line `s = linspace(0, halfSpan, N+1)';` creates 91 stations for 90 intervals. `linspace` includes both endpoints; `'` makes a column vector. In the Command Window, enter `size(s)`, `s(1)`, and `s(end)`. Verify size 91-by-1 and endpoints 0 and 0.450 m.
2. **Calculate dynamic pressure.** Replace `q = NaN` using $q = \frac{1}{2} \rho V^2$. MATLAB uses `^` for a scalar power.
3. **Calculate chord at every station.** Translate $c_r + (c_t - c_r)s/s_{\max}$ into MATLAB with the supplied names `rootChord`, `tipChord`, and `halfSpan`. Check `c(1)` and `c(end)` against the input table.
4. **Activate only the outer region.** Use `max(0, (s-activation)/(halfSpan-activation))` for `g`. Elementwise comparison with zero removes negative values inboard. This expression is used only on the supplied grid $0 \leq s \leq s_{\max}$. Check `min(g)`, `max(g)`, and `g(s<activation)`.
5. **Calculate both load arrays.** Use the force laws in the Part 2 model definition. The product of two station arrays needs `.*`, for example `c .* g`. Ordinary `c*g` is matrix multiplication and is not the operation required here. Use `./` for elementwise division and `.^` for elementwise powers if needed.
6. **Build the two moment integrands.** Fill `mxDensity` with `s .* fz` and `mzDensity` with `-s .* fx`. Their units are N; integrating over metres gives N·m.
7. **Apply the trapezoidal rule.** The syntax is `trapz(stations, values)`. Fill `Fz = trapz(s, fz);`, then apply the same pattern to `fx`, `mxDensity`, and `mzDensity`. **Do not use `trapz(fz)`**: that assumes station spacing of one rather than 0.005 m.
8. **Run Section 2C.** When your values are finite, it displays the result table, saves station data using `table` and `writetable`, and draws three plots using `subplot` and `plot`. Read each axis label and explain the zero-load region. Open the output folder printed by the setup.

A small completed example of `trapz`

```
s_demo = [0; 0.2; 0.4]; % m
f_demo = [0; 3; 6];      % N/m
F_demo = trapz(s_demo, f_demo);
```

The two trapezoids are $0.2(0 + 3)/2 = 0.3$ N and $0.2(3 + 6)/2 = 0.9$ N, so MATLAB returns 1.2 N. Your assessed calculation uses the same procedure with 90 intervals.

Check sensitivity to grid spacing and speed

2C-1. Refine the grid [5 points]

1. Keep the baseline $V = 15$ m/s. In starter **Section 2D**, sFine contains 181 stations for $N = 180$ intervals.
2. Recompute cFine, gFine, fzFine, and fxFine on sFine. Copy the physical laws, then replace the coarse variable names with their fine-grid counterparts. Do not combine 181 station locations with 91 force values.
3. Store the four fine-grid integrals in a row in this order: [DeltaFz, DeltaFx, DeltaMx, DeltaMz]. In the starter this row is named fine; the corresponding coarse row is baseline.
4. For each quantity Q , calculate

$$\text{relative change (percent)} = 100 \frac{|Q_{180} - Q_{90}|}{|Q_{180}|}.$$

Use $100 \cdot \text{abs}(\text{fine} - \text{baseline}) / \text{abs}(\text{fine})$. All four reference magnitudes are nonzero for this model. A change below **0.1%** is adequate here. State whether each result passes. Grid agreement checks numerical resolution; it does not validate the aerodynamic model.

2C-2. Predict and test speed scaling [5 points]

1. Before running starter **Section 2E**, predict the ratio of every force and moment at 10 and 20 m/s relative to 15 m/s. Write your prediction as $(V/15)^2$ and evaluate it.
2. The supplied loop is for $j = 1:\text{numel}(\text{speeds})$, where $\text{speeds} = [10;15;20]$. At each iteration, use $\text{speeds}(j)$ to calculate a new dynamic pressure q_j .
3. Recompute fzj and fxj using q_j and the coarse $N = 90$ geometry. Compute all four integrals again. Store the row in $\text{speedResults}(j,:)$; the colon means “all four columns.”
4. Make a table of speed and the four results. Divide each 10 and 20 m/s result by the matching 15 m/s result. Compare with the ratios predicted in Step 1. Keep geometry and coefficients fixed during this experiment.
5. Explain in two sentences why the results scale with V^2 **within this model** and which assumption could make that scaling fail for a real control device.

Checks you can make without an answer key

Inboard of 0.300 m both load increments should be zero. Changing speed must not change the chord or activation curves. The resultant forces should retain their predicted signs. Because the two load laws share the same spatial shape, the factor ratio A_z/A_x also constrains the ratios $\Delta F_z/(-\Delta F_x)$ and $\Delta M_x/\Delta M_z$; calculate and explain this check.

Submit from MATLAB: labeled load plots; the four baseline results with units; a four-row $N = 90$ versus $N = 180$ convergence table; and the speed table with predicted and computed ratios. Include your completed script, not screenshots of code alone.

Build the trapezoids in the WingLoads sheet

Use the supplied **WingLoads** sheet. Inputs are in B5:B13; B14 is for dynamic pressure. Keep $V = 15$ m/s and $N = 90$ for the submitted baseline. There are 91 station rows, **21:111**; A is station index, B is s in metres. A/B are already filled.

Use the Excel input-cell map in the Part 2 model definition (page 4); fixed input references use dollar signs.

1. Click **B14** and type $=0.5*B5*B6^2$. Press Enter. Confirm the result is positive and has units Pa.
2. Enter the following formulas in row 21, then select **C21:H111** and use **Home** → **Fill** → **Down** (or Ctrl+D). Absolute references such as \$B\$8 remain fixed; B21 changes with the station row.

Cell	Quantity	Formula
C21	Chord	$=\$B\$8+(\$B\$9-\$B\$8)*B21/\$B\7
D21	Activation	$=\text{MAX}(0,(B21-\$B\$10)/(\$B\$7-\$B\$10))$
E21	Δf_z	$=\$B\$14*C21*\$B\$11*D21$
F21	Δf_x	$=-\$B\$14*C21*\$B\$12*D21$
G21	Roll integrand	$=B21*E21$
H21	Yaw integrand	$=-B21*F21$

3. Rows represent **stations**; a trapezoid spans two adjacent stations. Leave the supplied “n.a.” entries in I21:L21 unchanged because row 21 has no preceding interval. Enter the first interval’s formulas in row 22:

Cell	Trapezoid contribution	Formula
I22	ΔF_z	$=(B22-B21)*(E22+E21)/2$
J22	ΔF_x	$=(B22-B21)*(F22+F21)/2$
K22	ΔM_x	$=(B22-B21)*(G22+G21)/2$
L22	ΔM_z	$=(B22-B21)*(H22+H21)/2$

4. Fill **I22:L111** down. Sum I22:I111 in **E5**, J22:J111 in **E6**, K22:K111 in **E7**, and L22:L111 in **E8**. Example: $=\text{SUM}(I22:I111)$. Include all 90 intervals.
5. Make a four-row comparison table in your report: quantity, MATLAB $N = 90$, Excel, absolute difference. Agreement within 10^{-9} N for forces and 10^{-9} N·m for moments is adequate; retain full precision during computation. Copy the full-precision MATLAB baseline printed by the script (or the exported Q2_baseline_totals.csv) into F5:F8. Enter $=E5-F5$ in G5 and fill through G8. Report ABS(G5) and its counterparts as absolute differences. Do not copy rounded display values.
6. Create one Excel **XY Scatter with lines** chart using B21:B111 as numerical x and E21:E111 as y. Label axes with units. Do not use a category-axis Line chart. Save the workbook with formulas visible when a cell is selected.

If your Excel uses semicolons between function arguments, replace the comma in MAX accordingly. Changing B13 alone does not create extra station rows; grid refinement is assessed in MATLAB.

Does an outboard side force necessarily roll the aircraft?

Notebook route: return to **7A-3b**, especially **Figure 7A-3b-3 — Same side force, different lines of action**. The lecture's numbers are different from the assessed values below.

Apply the same force $\mathbf{F} = (0, 2, 0)$ N at each location relative to the CG:

Case	x (m)	y (m)	z (m)	Before calculation
A	0	0.35	0	Does the force line pass through the CG?
B	-0.15	0.35	+0.04	Aft of and below the CG
C	-0.15	0.35	-0.04	Aft of and above the CG

3A. Sketch and predict [6 points]

- For each case, draw a **plan view and rear view**. Mark the CG, positive axes, application point, force arrow, and the extended line of action. Sketches may be schematic; write the actual offsets beside them.
- Without multiplying numbers, predict which moment components are zero and the signs of the nonzero components. State which component should reverse between B and C and why.

3B. Calculate three ways [9 points]

- Use the three component formulas in the coordinate-convention section. Show the substituted numbers for **Case B** by hand, including units; tabulate A/B/C.
- In MATLAB starter **Section 3**, the rows of rCases are A/B/C. In the loop, rCases(j,:) selects the current row. Replace the TODO with cross(rCases(j,:), Fside). Compare all components against the hand calculation.
- In Excel **Controls**, rows **7:9** contain the same cases. Columns B/C/D hold x/y/z; E/F/G hold $F_x/F_y/F_z$. Enter the formulas below in row 7 and fill through row 9.

Cell	Component	Formula
H7	M_x	=C7*G7-D7*F7
I7	M_y	=D7*E7-B7*G7
J7	M_z	=B7*F7-C7*E7

3C. Explain the geometry [5 points]

Answer in your own words: (i) Why is the outboard distance $y = 0.35$ m not sufficient by itself to create roll from this force? (ii) Which perpendicular arm controls roll and which controls yaw? (iii) Give one application point that produces nonzero yaw but zero roll for this same force. Verify your proposed point with cross.

Submit: six small view sketches; a prediction table; the hand calculation for B; a three-case moment table checked in MATLAB and Excel; and the three geometric explanations. The zero-moment case still has a nonzero resultant force.

Two surfaces share pitch and roll commands

Notebook route: 7A-11. What does each modern effector actually move or change? Read the generic elevon mixer, then try the following “7A-11. Elevon mixer — predict before running” code cell. Use it to understand commands; the extra force model below is supplied for HW2.

Positive local deflection means **trailing edge up** on each side. Let P request nose-up pitch and R request right-wing-down roll, in this instructional configuration:

$$\delta_L = P - R, \quad \delta_R = P + R, \quad -15^\circ \leq \delta_{L,R,\text{limited}} \leq 15^\circ.$$

Case	P (deg)	R (deg)	What to look for
1	8	0	Common-mode command
2	0	5	Differential command
3	8	5	Combined command
4	12	8	A command may exceed the actuator limit

4A. Predict surface motion [4 points]

Sketch left and right trailing-edge motion for each case **before running the notebook or MATLAB**. Calculate requested δ_L, δ_R . Mark any value outside $\pm 15^\circ$. A negative deflection means trailing edge down. These are surface commands, not aircraft pitch or bank angles.

4B. Convert commands to force increments [8 points]

For this exercise only, use the linear gain $k = 0.05$ N/deg and application points:

$$\mathbf{r}_L = (-0.15, -0.35, 0) \text{ m}, \quad \mathbf{r}_R = (-0.15, +0.35, 0) \text{ m},$$

$$\Delta \mathbf{F}_L = (0, 0, k\delta_L), \quad \Delta \mathbf{F}_R = (0, 0, k\delta_R).$$

Evaluate these force vectors twice: first using the requested deflections and then using the limited deflections. Positive $\Delta \mathbf{F}_z$ is a downward force increment. Because both force application points lie aft of the CG, the signs are consistent with the teaching mixer.

$$\Delta \mathbf{M}_{\text{total}} = \mathbf{r}_L \times \Delta \mathbf{F}_L + \mathbf{r}_R \times \Delta \mathbf{F}_R.$$

Find all three components; do not omit M_z simply because you expect it to vanish. Show one complete hand calculation for **Case 3 before limiting** and check all four cases with MATLAB and Excel.

What this model includes. A fixed linear force-per-degree gain, two fixed force application points, and symmetric actuator limits. It omits changes in drag, nonlinear aerodynamics, and aircraft motion. It is not the flight-control law of any named UAV, and a zero yaw result here does not establish zero yaw coupling for real elevons.

Apply the limit to each surface separately

MATLAB: complete Sections 4A and 4B

1. The rows of commands are the four cases. In the supplied loop, P and R are already read from the current row. Set requested(j,:) to [P-R, P+R].
2. Use min(max(requested(j,:),-limit),limit) to obtain the two limited deflections. The inner max prevents a value below -15; the outer min prevents a value above +15. Test the expression on [-18, 7, 21] separately so you understand the clipping.
3. Create row force vectors [0,0,k*delta] for each side, using the matching requested or limited deflection. Compute cross(rLeft,FLeft) plus cross(rRight,FRight). Store the requested and limited moment rows separately.
4. Recover the two attained command channels using the formulas below. The starter then exports a table and draws a grouped bar chart for Case 4 using bar(...,'grouped'). The two rows of the bar-chart data correspond to the two surfaces.

$$P_{\text{attained}} = \frac{\delta_{L,\text{limited}} + \delta_{R,\text{limited}}}{2}, \quad R_{\text{attained}} = \frac{\delta_{R,\text{limited}} - \delta_{L,\text{limited}}}{2}.$$

Read the output: requested and limited angles are in degrees; force increments are in newtons; moments are in N·m. The recovered P/R values are mixed commands, not measured attitude angles. Compare Case 4 before and after clipping without changing any geometry or gain.

Compare requested and attainable moments

Excel: Controls sheet, rows 22:25 and 45:48

B14 is the limit, B15 the gain, B16:B18 the left x/y/z coordinates, and E16:E18 the right x/y/z coordinates. B/C in each case row contain P/R. For the main table, enter row 22 formulas and fill through row 25. For the comparison table (header in row 44), enter row 45 formulas referencing Case 1 in row 22, then fill through row 48:

Cell	Meaning	Formula
D22 / E22	Requested left / right	=B22-C22 / =B22+C22
F22 / G22	Limited left / right	=MIN(MAX(D22,-\$B\$14),\$B\$14) =MIN(MAX(E22,-\$B\$14),\$B\$14)
H22 / I22	Limited $\Delta F_{z,L/R}$	=\$B\$15*F22 / =\$B\$15*G22
J22	Limited total M_x	=\$B\$17*H22+\$E\$17*I22
K22	Limited total M_y	=-\$B\$16*H22-\$E\$16*I22
L22	Limited total M_z	Enter the formula implied by the force vectors.
B45	Requested total M_x	=\$B\$17*\$B\$15*D22+\$E\$17*\$B\$15*E22
C45	Requested total M_y	=-\$B\$16*\$B\$15*D22-\$E\$16*\$B\$15*E22
D45	Requested total M_z	Enter the formula implied by the force vectors.
E45 / F45	Attained P / R	=(F22+G22)/2 / =(G22-F22)/2

4C. Explain saturation [8 points]

For Case 4, compare requested versus limited deflections, attained P/R channels, and M_x/M_y . Explain why limiting one physical surface changes **both** command channels. Which cases should remain unchanged by the limit? Explain why the attained channels are not measured aircraft attitude angles. Submit your prediction sketches, one hand calculation, result table, Case 4 chart, and explanation.

Follow the reactions into the airframe

Notebook route: 7A-1. Wing structure, 7A-5. Predict the forces; audit the load path, and 7A-14. Bring the control mechanism back to Aerospace Structures. Use the wing-anatomy figures to identify the physical members receiving each reaction.

5A. Draw two load-path sketches [5 points]

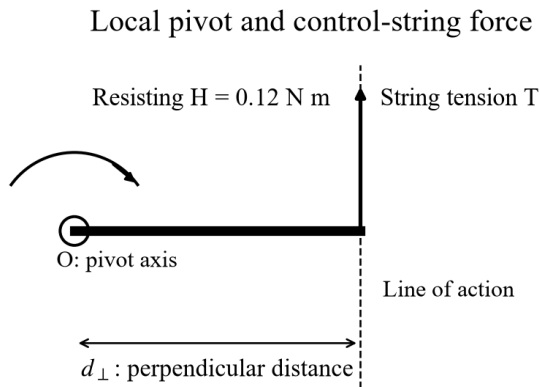
Draw one **modern upper-surface spoiler** and one **vertically pivoting wing-tip vane**. On each sketch:

1. Identify the moving part, its hinge or pivot, and the nearby wing structure. Show an aerodynamic force arrow acting on the moving part.
2. Draw the load route from the moving part through its hinge/bracket or pivot/post into local structure, then toward the wing root.
3. Add the **separate actuation branch**: actuator and anchor for the spoiler; string/control rod and its support for the vane. Label the reaction where that branch attaches to the structure.
4. Circle one connection that could require local reinforcement. State what additional information you would need to calculate its stress, such as geometry, material properties, or reaction distribution.

Use arrows and labels, not an unannotated aircraft photo. Avoid sending all load through the actuator alone or through the hinge alone without drawing the other branch.

Sketch planning space. Lay out the aerodynamic-load and actuator-reaction branches for both mechanisms. Label the local structure receiving each branch.

Use the perpendicular moment arm



Two paths carry the control loads

Surface load → panel → pivot / bracket

String tension → linkage / actuator anchor

Both paths connect to supporting structure

Ideal, frictionless mechanism in static balance. H is a supplied local hinge moment, separate from the aircraft roll moment.

5B. Calculate string tension from a supplied local torque [3 points]

A simple vane mechanism must resist a **local** torque $H = 0.12 \text{ N}\cdot\text{m}$ about its pivot. A string supplies balancing torque through a perpendicular moment arm $d_{\perp}$. Assume static torque balance and neglect friction for this calculation:

$$T = \frac{H}{d_{\perp}}, \quad d_{\perp} = 0.010, 0.020, 0.030 \text{ m.}$$

1. Sketch the pivot, string-force line, and the **perpendicular distance** between them. The perpendicular distance is not necessarily the attachment radius.
2. In MATLAB starter **Section 5**, replace the TODO with H / d . Explain why elementwise division gives one tension for each arm.
3. In Excel **Controls**, H is in **B30**; moment arms are in **B34:B36**. Enter $=B\$30/B34$ in **C34** and fill through C36. Compare with MATLAB and report tension in newtons.
4. Explain what happens to tension when the perpendicular arm is doubled, keeping H fixed.

5C. Distinguish two moments [2 points]

Can you take the aircraft's global rolling moment from Part 2 and automatically use it as H in the string-tension equation? Explain using the different reference axes and free bodies. Name one additional measurement or geometric detail needed to connect a device's aerodynamic load to its local resisting torque.

Bridge to your printed wing: identify one location where a moving device would introduce a new fitting or reinforcement requirement. A labeled conceptual sketch is sufficient; this homework does not add a moving-control mechanism to the scheduled fabrication deliverables.

Make your result easy to reproduce

Submit your individually authored **Surname_HW2_Report.pdf**, **Surname_HW2.m**, and **Surname_HW2.xlsx**, together with the course-required **Claim–Evidence–Check slide (.pptx and .pdf)** and a short **README**. Submit through Canvas; keep graded work out of the public repository. Include the plots and key tables in the PDF so a reader can follow your reasoning without opening the software. Keep full computation precision; round only displayed results to a sensible number of significant figures.

Part	Evidence required in your report	Points
1	Hardware labels; six pairs of paper coefficients with reading tolerances; coupling explanation; two-UAV comparison	15
2	Predictions and sketches; baseline plots/results; grid-refinement table; speed-scaling check; MATLAB–Excel comparison	35
3	Six view sketches; predicted signs; one hand calculation; three-case table; geometric explanations	20
4	Surface-motion predictions; one hand calculation; requested/limited force–moment results; saturation chart and explanation	20
5	Two structural load paths; local torque sketch and tension table; global/local moment distinction	10

Individual reflection and team cross-check record

Add a short paragraph naming your collaborators and recording one cross-check you performed independently. Then, in about **150–200 words**, explain (i) a force/moment sign you initially found difficult, (ii) how a sketch or independent calculation resolved it, and (iii) one result that depends on a simplifying model. This reflection supports the reasoning marks above; it is not an extra calculation problem or a separate team report.

Before submitting

1. Save, close, and reopen your MATLAB file. Run it from the top. Confirm that no assessed variable remains NaN, all result tables appear, and files are saved in hw2_outputs next to the script.
2. Open the Excel workbook and click representative cells from each part. Confirm that they contain formulas referencing the intended inputs. Verify the baseline input speed is restored to 15 m/s.
3. Check that every plot has a descriptive title or caption and labeled axes with units. Check that every moment table states its reference point and coordinate convention.
4. Read every prediction beside its computed result. If your prediction was wrong, explain the correction; do not silently remove the original reasoning.
5. Ensure you distinguish historical measurements, figure-reading estimates, and the supplied educational force laws. Include source references for copied figures.

Finish the standard homework portfolio

The question points organize the technical tasks. Final evaluation follows the established course rubric: **engineering judgment 40%; mechanics and numerical correctness 25%; independent verification 20%; one-slide communication 10%; reproducibility, attribution and AI disclosure 5%**. These are the normal course requirements, not extra problem points.

Engineering Judgment Record: use Part 2

Designate a short subsection of your report as “**Part 2 — Engineering Judgment Record.**” Reuse the work you already completed: state the system, prescribed loads, assumed application line, prediction, model assumptions and limitations; show at least two independent checks. Suitable checks include dimensional consistency, the MATLAB–Excel comparison, a limiting case, and the speed-scaling check. Distinguish numerical agreement from validation of the force law. No new calculation is needed.

One Claim–Evidence–Check slide

Create one slide in PowerPoint and export the same slide as PDF. Use one result from this homework. Include a clear claim, a readable figure or numerical result supporting it, an independent check, your confidence, and the principal limitation. Synthesize the reasoning rather than pasting a collage of screenshots. Follow the existing Canvas rotation for oral checks; this handout does not assign an additional presentation.

README and process / AI-use note

In a short README text file, state your MATLAB version, Excel version, filenames, how to run the script, and where the outputs appear. Add a process and AI-use note to the report or README: tools used, their contributions, changes you made, errors found, and how you verified the result. If applicable, write “**No generative AI used.**” Identify collaborators and sources. Keep your original predictions and explain any corrections.

If something goes wrong

Symptom	First check
MATLAB matrix-size error	Use column station arrays; inspect size; use .* for stationwise products.
Force is off by a factor related to grid spacing	Pass both arguments to trapz(s,f); include interval width in every Excel trapezoid.
Signs disagree between methods	Use cross(r,F); check +z down and the minus sign in the yaw integrand.
Excel formulas change input references when filled	Use dollar signs for fixed cells; allow station row references to change.
Excel matches MATLAB but units are wrong	Numerical agreement is insufficient: distinguish N/m, N, and N·m.
Only Case 4 gives unexpected coupling	Inspect each surface’s requested and limited angle before recomputing force and moment.

MATLAB function references: [trapz](#), [cross](#), [linspace](#), and [exportgraphics](#) (optional figure export in R2020a or later). The supplied starter uses print to save its PNG plots.